

Selcuk Kilinc Postdoctoral Researcher

Catskill 143 at School of Education | University at Albany skilinc@albany.edu +1 979 721 3467 Male LinkedIn
 ResearchGate Google Scholar 0000-0001-8846-7243 Personal Website

Postdoctoral researcher in science and STEM education specializing in AI-supported teaching and learning, with a primary research focus on AI literacy, generative AI, and mixed-methods studies of how AI tools support inquiry, reflection, and pedagogical decision-making.

EDUCATION

- Doctor of Philosophy / Science Education**02/2018 – 07/2023 | Turkey
Middle East Technical University
 - Dissertation:** Real-Time Air Quality Monitoring as A Catalyst for the Growth of Pre-Service Science Teachers in Citizen Science
- Master of Science / Software Engineering**09/2020 – 07/2022 | Turkey
Middle East Technical University
 - Graduation Project:** SmartCal: Innovating Mobile Tech for Real-Time Calorie Tracking and Dietary Guidance
- Master of Science / Secondary Science and Mathematics Education**01/2015 – 01/2018 | Turkey
Middle East Technical University
 - Thesis:** Exploring of STEM Readiness of a Faculty of Education in Turkey
- Integrated Bachelor's and Non-Thesis Master's Chemistry Education**09/2007 – 06/2014 | Turkey
Middle East Technical University
 - Graduation Project:** Goal Orientations and Task Values of High School Students

PROFESSIONAL EXPERIENCE

- Postdoctoral Research Scholar**01/2026 – Present | NY, USA
University at Albany, State University of New York
 - Conducting research on artificial intelligence in education, with a focus on AI-supported teaching and learning in higher education.
 - Examining how generative AI tools are conceptualized and used in educational contexts, including implications for learning and assessment.
 - Supporting scholarly publications, grant-related activities, and interdisciplinary collaborations in AI in education, learning sciences, and edtech.
- Postdoctoral Research Associate**01/2025 – 12/2025 | TX, USA
Texas A&M University
 - Co-designed and implemented an AI-enhanced Ag-STEM program across two rural school districts, reaching 80+ middle school students.
 - Integrated AI chatbot tutors and Arduino-based IoT sensors to scaffold computational thinking and inquiry in classroom instruction.
 - Led teacher professional development and classroom support, contributing to grant proposals, invited lectures, and peer-reviewed publications.
- Middle East Technical University Faculty of Education**

Research & Teaching Assistant09/2015 – 12/2024 | Turkey
 - Conducted research, published papers, and presented at international conferences in science education and educational technology
 - Designed and taught undergraduate and graduate courses in instructional technology, research methods, and science pedagogy
 - Developed AI- and IoT-enhanced learning environments and mentored students in qualitative, quantitative, and mixed-methods research.

Coordinator for Research Assistants
 - Led 20+ assistants, strategically aligning their responsibilities with faculty priorities and coordinating across departmental initiatives.
 - Implemented efficient scheduling and workload distribution systems, ensuring the timely completion of research and instructional duties.

Distance Education Coordinator
 - Helped lead the faculty's transition to online education during COVID-19, supporting 40+ faculty and 200+ students across the faculty.
 - Collaborated with instructors to develop interactive online courses and used learning analytics to inform improvements.

Technology Coordinator
 - Provided technical support and training on educational technologies for 40+ faculty members and staff.
 - Supported the redesign and maintenance of the department website, improving navigation and access to key information.
- Chemistry Teacher**09/2014 – 08/2015 | Turkey
Jale Tezer High School
 - Planned and delivered curriculum-aligned chemistry instruction for 300+ high school students with hands-on experiments.
 - Designed and implemented online educational materials to support differentiated instruction.

HONORS & AWARDS

- TUBITAK 2219 International Postdoctoral Research Fellowship Program**2025
The Scientific and Technological Research Council of Turkey
 - Awarded a competitive postdoctoral research fellowship (\$23,400) to research AI-enhanced STEM education at Texas A&M University, USA.
- TUBITAK 2224-A – International Conference Travel Grant**2025
Overseas Scientific Meetings Participation Support Program
 - Received approximately \$1,700 in travel and registration support to present research at the NARST 2025 Annual International Conference.
- Dean's Honor List Student**2013
Middle East Technical University (METU)
 - Placed on the university honor list twice (January 2013 and July 2013) in recognition of high academic achievement during BS studies.

Peer-Reviewed Journal Articles

- Aldemir T., Bicer A., **Kilinc S.**, Moon J. & Kwok, M. (2026). Challenges, Safeguards, and Professional Learning Needs for AI Integration: Insights from a Two-Week AI Literacy Module with Preservice Teachers. *Cogent Education*. (accepted for publication)
- Evkaya, O., **Kilinc, S.**, & Kizilates, S. (2026). Cross-national perceptions of generative AI in higher education: A comparative study of university students in the UK and Turkey. *International Journal of Information and Education Technology*, 16(3), 640–647. <https://doi.org/10.18178/ijiet.2026.16.3.2536>
- Aldemir T., **Kilinc S.**, Bicer A., Grant P., Davis, T. J. & Sweany, N. W. (2025). Intelligent-TPACK in Practice: Design and Evidence from a Three-Week Teacher Preparation Module. *Computers and Education Open*, 9, 100306. <https://doi.org/10.1016/j.caeo.2025.100306>
- Aldemir T., Bicer A., **Kilinc S.**, Moon J., & Kwok, M. (2025). Exploring Emergent AI-TPACK Competencies in a Two-Week AI Literacy Module for Preservice Teachers. *Teaching and Teacher Education*, 168, 105231. <https://doi.org/10.1016/j.tate.2025.105231>
- Kilinc, S.**, Geban, O. & Ozturk, G. (2025). STEM in Teacher Education: Readiness, Challenges, and Alignment with Global Frameworks. *Ankara University Journal of Faculty of Educational Sciences (JFES)*, 58(3), 981-1032. <https://doi.org/10.30964/auebfd.1567788>
- Kilinc, S.** (2024). Comprehensive AI assessment framework: Enhancing educational evaluation with ethical AI integration. *Journal of Educational Technology and Online Learning*, 7(4-ICETOL 2024 Special Issue), 521-540. <https://doi.org/10.31681/jetol.1492695>
- Kilinc, S.** (2023). Embracing the Future of Distance Science Education: Opportunities and Challenges of ChatGPT Integration. *Asian Journal of Distance Education*, 18(1), 205-237. <https://doi.org/10.5281/zenodo.7857396>
- Uzuntiryaki-Kondakci, E., Tuysuz, M., Sarici, E., Soysal, C., & **Kilinc, S.** (2021). The role of the argumentation-based laboratory on the development of pre-service chemistry teachers' argumentation skills. *International Journal of Science Education*, 43(1), 30-55. <https://doi.org/10.1080/09500693.2020.1846226>

Peer-Reviewed Book Chapters

- Kilinc, S.** (2025). Personalizing Education in the AI Era: The Comprehensive Impact of Customized Chatbots Across Educational Domains. In *Artificial Intelligence and Human Agency in Education: Volume Two: AI for Equity, Well-Being, and Innovation in Teaching and Learning* (pp. 27–65). Springer. https://doi.org/10.1007/978-981-96-9251-4_2

Peer-Reviewed Proceedings

- Kilinc, S.**, Kumi-Yeboah, A., & Feyzi Behnagh, R. (2026). Beyond plagiarism: Prompt leakage, sensitive data exposure, and misinformation in educational GenAI. In *Proceedings of the 21st Annual Symposium on Information Assurance (ASIA '26), in conjunction with the 28th New York State Cyber Security Conference (NYSCSC)*. (accepted for publication in proceedings).
- Kilinc, S.**, Aldemir, T., Sabanwar, V., Bicer, A., & Song, D. (2026). Planners and tinkerers: A comparative case study of K-12 students' engagement with data practices in an Ag-STEM. In *Proceedings of the International Conference of the Learning Sciences (ICLS 2026)*. International Society of the Learning Sciences. (accepted for publication in proceedings).
- Sabanwar, V., Aldemir, T., **Kilinc, S.**, Bicer, A., & Song, D. (2026). How middle school students engage in computational literacies in an agricultural class. In *Proceedings of the International Conference of the Learning Sciences (ICLS 2026)*. International Society of the Learning Sciences. (accepted for publication in proceedings).
- Aldemir, T., **Kilinc, S.**, Sabanwar, V., Bicer, A., & Song, D. (2026). Disciplinary computational thinking in AI-mediated problem framing: A rural Ag-STEM context. In *Proceedings of the International Conference of the Learning Sciences (ICLS 2026)*. International Society of the Learning Sciences. (accepted for publication in proceedings).
- Kilinc, S.**, Aldemir, T., Sabanwar, V., Bicer, A., & Song, D. (2025). AI Chatbot Coaching for Elevating Student Research. In *Proceedings of the 2025 ACM Conference on International Computing Education Research V. 2* (pp. 1–2). Association for Computing Machinery. <https://doi.org/10.1145/3702653.3744315>

Manuscripts Under Review

- Kilinc S.**, Aldemir T., Misiejuk K., Kaliisa R., Bicer A., Sabanwar, V., Song, D., Yalvac B. (2026). A Mixed-Methods Analysis of AI Scaffolding Patterns and Student Inquiry Profiles in a Middle-School Agriculture-STEM Classroom. *Computers & Education: Artificial Intelligence*. (Journal Article)
- Kilinc, S.**, Feyzi-Behnagh, R., & Kumi-Yeboah, A. (2026). Mapping trust boundaries in AI-mediated environments: A behavioral cybersecurity framework. In *Artificial Intelligence in Behavioral Cybersecurity*. World Scientific Publishing. (Book Chapter)

PRESENTATIONS

Peer-Reviewed Conference Presentations

- Kilinc, S.**, Kumi-Yeboah, A., & Feyzi Behnagh, R. (2026). *Beyond plagiarism: Prompt leakage, sensitive data exposure, and misinformation in educational GenAI*. Presented at the 21st Annual Symposium on Information Assurance (ASIA '26), in conjunction with the 28th New York State Cyber Security Conference (NYSCSC).
- Kilinc, S.**, Aldemir, T., Sabanwar, V., Bicer, A., & Song, D. (2026). *Planners and tinkerers: A comparative case study of K-12 students' engagement with data practices in an Ag-STEM*. Accepted for presentation as Long Paper at the International Conference of the Learning Sciences (ICLS) 2026.
- Sabanwar, V., Aldemir, T., **Kilinc, S.**, Bicer, A., & Song, D. (2026). *How middle school students engage in computational literacies in an agricultural class*. Presented at the International Conference of the Learning Sciences (ICLS) 2026.
- Aldemir, T., **Kilinc, S.**, Sabanwar, V., Bicer, A., & Song, D. (2026). *Disciplinary computational thinking in AI-mediated problem framing: A rural Ag-STEM context*. Presented at the International Conference of the Learning Sciences (ICLS) 2026.
- Kilinc, S.**, Yilmazoglu, E., Inceci, I., & Kondakci, E. (2026). *From Tool to Teammate: Reimagining Student-AI Collaboration in Engineering Design-Based STEM*. Presented at the National Association for Research in Science Teaching (NARST) Annual International Conference, Seattle, WA, USA.
- Kondakci, E., Inceci, I., Yilmazoglu, E., & **Kilinc, S.** (2026). *Enhancing High School Students' Self-Regulation with an AI-supported Engineering Design Process*. Presented at the National Association for Research in Science Teaching (NARST) Annual International Conference, Seattle, WA, USA.
- Kilinc, S.**, Aldemir, T., Sabanwar, V., Bicer, A., & Song, D. (2026). *The AI as Cognitive Mentor: Scaffolding Student Scientific Inquiry Through Dialogue*. Presented at the American Educational Research Association (AERA) Annual Meeting, Los Angeles, CA, USA.
- Kilinc, S.**, Aldemir, T., Sabanwar, V., Bicer, A., & Song, D. (2026). *Computational Thinking and Participation in Rural Ag STEM: A Comparative Case Study of Two Middle School Teams*. Presented at the American Educational Research Association (AERA) Annual Meeting, Los Angeles, CA, USA.
- Sabanwar, V., Aldemir, T., **Kilinc, S.**, Bicer, A., & Song, D. (2026). *Teacher Conceptualization of Student Challenges and Solutions for CT in A Non-CS Classroom*. Presented at the American Educational Research Association (AERA) Annual Meeting, Los Angeles, CA, USA.
- Evkaya, O., **Kilinc, S.** & Kizilates, S. (2025). *Can an AI-Powered Game-based Presentation Platform Be Used to Improve Students' Public Speaking Skills?*. Presented at the International Conference on Education and Artificial Intelligence Technologies (EAIT) Convention, Manchester, UK.

- **Kilinc, S.**, Aldemir, T., Sabanwar, V., Bicer, A., & Song, D. (2025). *Enhancing Research Questions and Hypothesis Development Through AI Chatbot Assistance in Rural Agricultural Education*. Presented at the Association of Educational Communications & Technology (AECT) International Convention, Las Vegas, NV, USA.
- Sabanwar, V., Aldemir, T., **Kilinc, S.**, Bicer, A., & Song, D. (2025). *Exploring Computational Thinking skills in the era of GenAI in K-12 education*. Presented at the Association of Educational Communications & Technology (AECT) International Convention, Las Vegas, NV, USA.
- **Kilinc, S.**, Aldemir, T., Sabanwar, V., Bicer, A., & Song, D. (2025). *AI Chatbot Coaching for Elevating Student Research*. Poster presented at the ACM Conference on International Computing Education Research (ICER), Charlottesville, VA, USA.
- **Kilinc, S.** & Ozturk, G. (2025). *Innovative Hybrid Science Education: Integrating Citizen Science & Digital Learning for Future-Ready Teachers*. Presented at the National Association for Research in Science Teaching (NARST) Annual International Conference, National Harbor, MD, USA.
- **Kilinc, S.**, & Ozturk, G. (2024). *Vatandas Bilimi Projelerinin Ogretmen Adaylarinin Bilimsel Yeterlilikleri Uzerindeki Etkisinin Degerlendirilmesi*. Presented at the National Science and Mathematics Education Congress, Edirne, Turkey.
- **Kilinc, S.** (2024). *Shifting Paradigms: Ethical Approaches and Integrity Guidelines for AI-Assisted Assessments in Learning*. Presented at the International Conference on Educational Technology and Online Learning (ICETOL), Eskisehir, Turkey.
- **Kilinc, S.**, Geban, O., & Ozturk, G. (2018). *Exploring Instructors' Perceptions of A Faculty Of Education In Turkey About STEM Education*. Presented at the Annual European Conference on Educational Research (ECER), Bolzano, Italy.
- **Kilinc, S.**, Tuysuz, M., Sarici, E., Soysal, C., & Kondakci, E. (2017). *Contribution of the argumentation-based laboratory to pre-service chemistry teachers' microscopic explanations of chemistry concepts*. Presented at the National Association for Research in Science Teaching (NARST) Annual International Conference, San Antonio, TX, USA.
- Soysal, C., Sarici, E., Tuysuz, M., **Kilinc, S.**, & Kondakci, E. (2017). *Development of Pre-service Chemistry Teachers' Argumentation Skills in Implementing Science Writing Heuristic at Chemistry Laboratory Subject/Problem*. Presented at the National Association for Research in Science Teaching (NARST) Annual International Conference, San Antonio, TX, USA.
- Ekiz Kiran, B., Kutucu, E. S., **Kilinc, S.**, Soysal, C., & Boz, Y. (2016). *Kimya Ogretmen Adaylarinin Kimya Bilgilerini Kullarak Gunluk Hayat Olaylarini Aciklama Duzeyleri*. Presented at the National Science and Mathematics Education Congress, Trabzon, Turkey.

Invited Lectures & Workshops

- From Questions to Insights: Demonstrating Qualitative Design in Action (2025). *Guest lecture for EDCI 690, Department of Teaching, Learning & Culture, Texas A&M University, USA.*
- Citizen-Science + IoT: Pathways to Authentic STEM & Future-Ready Teachers (2025). *Guest lecture for ENGR/SCIED 497F, Department of Teaching, Learning & Culture, Texas A&M University, USA.*
- Generative AI and Custom GPT Usage in STEM Projects (2025). *Invited lecture, TED Ankara College (Online), Ankara, Turkey.*
- Distance Education and LMS Integration into Teaching (2020 & 2021). *Workshop, Middle East Technical University, Ankara, Turkey.*

PROJECTS

Artificial Intelligence Mentorship for Rural Education

01/2025 – 01/2026 | Texas, USA

Educational Innovation Project

- Co-designed and implemented a 16-week AI-enhanced agriculture-STEM curriculum for 80+ middle school students, emphasizing computational thinking and data-driven environmental investigation.
- Integrated Arduino-based IoT sensors and a custom AI chatbot tutor to scaffold inquiry, troubleshooting, and analysis of authentic environmental data.
- Led curriculum co-design and teacher professional development, supporting implementation across Caldwell and Brenham Junior High Schools and generating data for conference presentations and journal manuscripts.

Permaculture Chatbot & Engineering-Design Mentorship

10/2024 – 09/2025 | Turkey

Educational Design Project

- Developed a chatbot to support permaculture learning and engineering-design mentorship, informed by a design-based research framework.
- Implemented an instructional intervention with ~70 students, including a two-day introductory unit and a subsequent two-month project phase where students used a custom chatbot to develop and iterate on permaculture design projects.
- Iteratively refined the system based on user feedback, interaction data, and emerging learner needs.
- Generated design insights into how AI-supported mentorship can scaffold ecological design thinking and problem-solving.

IoT-Enabled Citizen Science: Real-Time Air Quality Monitoring

09/2022 – 02/2023 | Turkey

Dissertation Project

- Designed and built WHO-aligned Arduino-based mobile air-quality monitor (PM_{2.5}, PM₁₀, key gas pollutants) with real-time display and data log.
- Integrated the device into a science methods course with 12 preservice teachers as an authentic citizen-science project.
- Supported preservice teachers in posing research questions, collecting air-quality data across diverse campus locations, and analyzing and visualizing the datasets.
- Facilitated data-informed presentations and recommendations to university leadership and documented the project in a short field video.

Determining of Education Faculties' STEM Readiness

01/2017 – 12/2017 | Turkey

Scientific Research Project

- Assessed the STEM readiness of education faculty members and preservice teachers after STEM education was introduced into the curriculum.
- Collected survey and interview data to identify needs in instructional technology use, interdisciplinary collaboration, and perceptions of STEM.
- Produced a policy report that informed Turkish Higher Education Council professional-development and curriculum-reform efforts.

TEACHING EXPERIENCES

U.S. Teaching & Professional Development

01/2025 – 12/2025 | TX, USA

Texas A&M University & Caldwell Junior High School

- Co-designed and taught a 16-week AI-enhanced agriculture-STEM program for 80+ junior high students, using co-teaching cycles and independently led lessons to support students' development of computational thinking and scientific reasoning.
- Modeled inquiry- and project-based STEM instruction, providing real-time scaffolding as students used AI chatbots and Arduino-based IoT sensors to investigate authentic agricultural problems.
- Designed and facilitated ongoing, job-embedded professional development and coaching for partner teachers on integrating AI tools and sensor data into their daily lessons and assessment practices.
- Mentored a PhD student in research design, data collection, validity and reliability, and academic writing for publication.

Instructor & Course Designer (Undergraduate & Graduate Courses)

- *SSME518 & MSE329 - Instructional Tech & Material Development (Face-to-Face & Hybrid, BS & MS)*
 - Led both graduate and undergraduate courses from concept to execution, including syllabus design, project-based learning activities, and assessment frameworks aligned with inquiry-based STEM education.
 - Designed curricula focused on developing IoT-based environmental monitoring projects using Arduino, enabling students to collect, analyze, and interpret real-world environmental data.
 - Pioneered the integration of Generative AI into the course, empowering students to create innovative, adaptive learning content.
- *MSE405 - Lab Applications in Science Education I (Face-to-Face, BS)*
 - Headed hands-on laboratory sessions for chemistry education, focusing on building students' critical thinking, experimental design, and scientific reasoning skills.

Teaching Assistant & Mentor (Graduate Courses)

- *MSE603, MSE602, MSE502 - Research Methods & Qualitative Research & Advanced Research (Face-to-Face & Online, MS & PhD)*
 - Provided mentorship to MS/PhD students in qualitative and quantitative educational research methods and research ethics.
 - Mentored undergraduate and graduate students in designing technology-enhanced science education projects, including AI-supported instructional tools and sensor-based inquiry activities.

Teaching Assistant & Mentor (Undergraduate Courses)

- *MSE409 & MSE410 - Practice Teaching I and II (Face-to-Face, BS)*
 - Supervised and mentored pre-service teachers during school-based practicum experiences, providing structured feedback on lesson planning, classroom instruction, micro-teaching sessions, and teaching portfolios to support reflective practice and professional growth.
- *MSE310 & MSE411 & MSE305 - Secondary Science Teaching Methods I and II & Assessment in Science Education (Face-to-Face, BS)*
 - Guided pre-service teachers in applying inquiry-based teaching methods and instructional strategies to design effective science lessons.
 - Guided pre-service teachers in designing and implementing assessment strategies, including formative and summative assessment, rubric development, and evaluation of student learning.

Teacher Mentoring & Instructional Design

09/2019 – 12/2024 | Turkey

Bestepe College & TED Ankara College

- Coached IB science teachers with professional developments including classroom observations, feedback conferences, collaborative planning.
- Facilitated workshops on inquiry-based & interdisciplinary curriculum design, student-centered assessment, and the use of digital and AI tools.
- Designed and delivered interactive sessions for students and teachers on the ethical and productive use of Generative AI to foster digital literacy.

K-12 Classroom & Intern Experience

09/2013 – 08/2015 | Turkey

Jale Tezer High School & Ankara Anatolian High School & METU High School

- Designed and delivered curriculum-aligned chemistry lessons for high school students, integrating hands-on laboratory experiments and digital resources to support diverse learners.
- Completed supervised teaching practicums in multiple high schools, collaborating with mentor teachers on lesson planning, lab management, and providing one-on-one academic support to students.

PROFESSIONAL SERVICE**Journal Reviewer**

- Computers & Education Open
- Teaching and Teacher Education
- Frontiers in Education
- Education Innovations: Systems and Future Learning (EISFL)
- Information and Learning Sciences
- AIS Transactions on Human-Computer Interaction
- International Journal of Science, Technology and Society
- Learning Letters
- Acta Infologica (ACIN)
- Turkish Journal of Educational Sciences
- AFJUR African Journals, Universities and Research

Conference Reviewer

- National Science and Mathematics Education Congress 2026
- Association for Educational Communications and Technology International Convention 2026
- International Conference on Artificial Intelligence in Education 2026
- International Conference of the Learning Sciences 2026
- International Conference on Computer-Supported Collaborative Learning Annual Meeting 2026
- Association for Educational Communications and Technology Online Conference 2026
- National Association for Research in Science Teaching Annual International Conference 2026
- American Educational Research Association Annual Meeting 2026
- National Association for Research in Science Teaching Annual International Conference 2025
- National Science and Mathematics Education Congress 2024

Conference Program Committee

- ISLS Annual Meeting 2025
- National Science and Mathematics Education Congress 2024
- 2026 NARST Annual International Conference

CERTIFICATES**AI & Programming & Engineering**

- Advanced AI App Design for Educators (PLC 201), Playlab.ai, 2025
- AI for Education Summit, AI for Education, 2024
- Career Essentials in Generative AI, Machine Learning, and Ethics, Microsoft & LinkedIn, 2023
- Python 3 Programming Specialization, University of Michigan (Coursera), 2022
- Complete Python Bootcamp: From Zero to Hero, Udemy, 2021
- Information Technologies Certificate (1-year program), METU CEC, 2021

Research Methodologies

- Instructional Design Foundations and Applications, University of Illinois Urbana-Champaign, 2025
- Data Science Methodology, IBM (Coursera – Alex Aklson & Polong Lin), 2022
- Qualitative Research, University of California, 2021
- Understanding Clinical Research: Behind the Statistics, Univ. of Cape Town, 2020
- Computer-Assisted Qualitative Data Analysis (MAXQDA certified training), 2019
- Methods & Statistics in Social Sciences Specialization, Univ. of Amsterdam, 2019

Leadership and Management

- Project Management & Career Development Specialization, University of California (Coursera), 2025
- Team & Project Management (32-week School of Leadership program), 2021
- Mentoring Program (8-month, Association for Coaching-accredited), 2019
- Creative Drama Leadership (1-year certificate), Contemporary Drama Association, 2014

SKILLS

Science Education & Learning Sciences

- Technology-enhanced K–12 and higher education science curricula design grounded in inquiry, modeling, and engineering design.
- Teacher professional development in science education and AI/EdTech integration, including workshop design, co-teaching, and ongoing coaching.
- Translating learning sciences and STEM education research into scalable classroom-ready tools, assessments, and PD resources.

Research Methodologies

- Mixed-methods research on technology-enhanced STEM and AI-in-education contexts.
- Qualitative approaches, including semi-structured interviews, focus groups, classroom observations, document and artifact analysis, and thematic/content analysis (MAXQDA).
- Quantitative approaches, including survey design and validation, scale development, experimental and quasi-experimental designs, and inferential statistics (t-tests, ANOVA, regression), plus instrument development for questionnaires, rubrics, and performance assessments.
- Learning analytics and multimodal data analysis, including log/clickstream and time-on-task data, video-based classroom interaction coding, and Epistemic Network Analysis (ENA).

AI & Instructional Technology & Software Engineering

- Design and evaluation of online, hybrid, and blended courses using instructional design models, TPACK, and AI-TPACK frameworks.
- Development of AI-driven tutoring systems, chatbots, and adaptive learning environments, with attention to ethics, feedback, and learner support.
- IoT-based learning environments using Arduino sensor networks and real-time data dashboards to support inquiry and data literacy.
- Programming for educational research and data systems (Python, basic Java) and rapid prototyping of research tools and pipelines.
- LMS and educational platforms (Canvas, Moodle, Google Classroom, etc.) for course design, analytics, and integration of external tools.

Collaboration & Leadership

- Mentoring K–12 teachers and preservice teachers in STEM, AI literacy, assessment, and classroom implementation of new tools.
- Coordinating interdisciplinary teams of educators, technologists, designers, and researchers across schools and universities.
- Competitive grant development, reporting, and multi-site project coordination with school districts and community partners.
- Designing & facilitating professional development workshops and co-design sessions in STEM and AI in education.

MEMBERSHIP IN PROFESSIONAL ASSOCIATIONS

The International Society of the Learning Sciences (ISLS)

National Association for Research in Science Teaching (NARST)

American Educational Research Association (AERA)

Association for Educational Communications and Technology (AECT)

COMMUNITY ENGAGEMENTS

ANKA Youth Association

2020 – present

Society Member

- Organized international youth projects under Erasmus+ (e.g., “EthiTech: Fostering Responsible Digital Citizenship in Youth” in Romania), recruiting participants and promoting digital literacy and intercultural dialogue.
- Coordinated Speaking Clubs (e.g., “AI & The Future”, “Is Money Key to Happiness”), facilitating discussions on AI, ethics, and societal issues.
- Supported hybrid internship programs (e.g., “Unlock Your Potential” with Jugendvision e.V., Germany), connecting Turkish youth to global opportunities in project management and digital media.
- Contributed to the “#DijitalKöprüler: Strengthening Intercultural Dialogue through Responsible Digital Citizenship” event (Antalya, Turkey), featuring workshops on digital ethics, online security, and intercultural exchange—aligning with the themes of my dissertation.

Turkish Intelligence Foundation

2018 – present

Volunteer Staff

- Contributed to the annual “Intelligence and Talent Congress” (2013–Present), assisting with logistics and facilitating sessions on intelligence, talent development, and educational innovation at METU.
- Supported Turkey Intelligence Foundations Meetings (animation science and healthy nutrition), engaging educators and students.
- Assisted with session planning, breakout workshops, and expert panel logistics addressing topics such as gifted education, innovative testing methods, and the role of social communication in talent development.

Darüşşafaka Society

2017 – present

Community Volunteer

- Participated in fundraising campaigns with companies and individuals to secure donations for underprivileged students' education.
- Served as a mentor in the society's mentorship program, guiding graduates in career planning and academic growth by one-on-one support.
- Facilitated student-professional networking events, connecting students with industry experts to promote educational equity.

METU Alumni Association

2016 – present

Member

- Contributed to various activities—including social, cultural, and professional development events—and actively participated in committees (e.g., Event, Communication, and Disaster & Emergency Assistance Committees).
- Coordinated and facilitated mentorship programs for numerous university students, providing career guidance and networking opportunities.
- Played a significant role in the February 2023 Turkey earthquake relief efforts by organizing donation drives, coordinating packaging, and ensuring efficient aid distribution to affected communities.
- Collaborated with cross-functional teams to enhance volunteer engagement and streamline logistical operations for big events.

The Educational Volunteers Foundation of Turkey

2013 – present

Education Volunteer

- Actively contributed to various fundraising events and donation campaigns that support underprivileged children's educational opportunities.
- Supported developing and disseminating engaging educational content through TEGV Akademi for children, parents, volunteers, and teachers.
- Participated in aid collection, packaging, and distribution activities, particularly during the February 2023 earthquake.

REFERENCES

Professional and academic references will be provided upon request.